R13

Code No: 126AQ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, December - 2018
INFORMATION SECURITY
(Computer Science and Engineering)

Time: 3hours **Max.Marks:75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

used paper 21-12-2018 Am

PART- A

(25 Marks)

1.a)	Compare transposition ciphers with substitution cipher.	[2]
b)	Explain the principles of security.	[3]
c)	What are the advantages of public key cryptography algorithm comparing ryman encryption algorithm.	[2]
d)	List the advantages of elliptic-curve cryptography.	[3]
e)	List three approaches to message authentication.	[2]
f)	What is the function of TGS server in Kerberoes.	[3]
g)	What are S/MIME message?	[2]
h)	List the different encryption and authentication algorithms used for AH and ESP protocols.	[3]
i)	What are the limitations of firewalls?	[2]
j)	What is intruder?	[3]

PART-B

(50 Marks)

2.a) Consider the following:
Plaintext: “KEY”
Secret key: “CRYPTOGRAPHY”
Compute the cipher text from given plain text and key using hill cipher method.
b) Explain the model for network security. [5+5]

OR

3.a) Explain the transposition techniques.
b) What are the advantages of steganography comparing with cryptography? [5+5]

4. Explain the AES algorithm. [10]

OR

5.a) Write short notes on key distribution.
b) In an RSA system, the public key of a given user is e=31, n=3599. What is the private key of this user? [5+5]

6. Explain whirlpool algorithm. [10]

OR

7. Explain X.509 authentication service. [10]

8. Explain the operation PGP message generation and message reception. [10]

OR

9.a) What are the cryptographic algorithms used in S/MIME?
b) Draw and explain fields in AH header. [5+5]

10. Explain secure inter branch payment transactions. [10]

OR

11.a) What is password management?
b) What are the various virus counter measures? [5+5]

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